**BMSE2073 SOFTWARE DESIGN AND ARCHITECTURE**
**ASSIGNMENT DESCRIPTION**

| | |
|---|---|
| **SEMESTER** | **202509** |
| **ASSESSED LEARNING OUTCOME** | CLO3 - Construct the software architecture for a project scenario. (P4, PLO3) |
| **SUBMISSION DEADLINE** | **Friday, 19 December 2025, 11.59 pm (Week 6)**<br><br>**University Regulation On Late Submission :**<br><br>*With the exception of Extenuating Mitigating Circumstance(EMC), the penalty for late submission of coursework shall be imposed after the submission deadline / extended submission deadline:*<br><br>• Late submission within 1 - 3 days: total marks to be deducted is 10 marks.<br>• Late submission within 4 - 7 days: total marks to be deducted is 20 marks.<br>• Late submission after 7 days: reject coursework and zero marks shall be awarded.<br><br>***Note:***<br>*For EMC (Extenuating Mitigating Circumstance) cases, students must apply to the FOCS office before the assessment deadline. Supporting documents, such as certified medical certificates, Death certificates, SPM examinations, MUET examinations, etc., must be attached to the application. The approval is subject to the application fulfilling the EMC requirements per university regulations.* |
| **ASSIGNMENT TASKS** | • For this assignment, group yourself into a **team of 4-5 members**. Students are **NOT** allowed to change their group without permission from the respective tutor.<br><br>• Please refer to **Appendix A** for the assignment task description.<br><br>• Students must do all required parts of the assignment and submit it on or before the defined deadlines (see ***Submission Deadlines section***).<br><br>• All students are reminded that this is their own individual group coursework. You are, therefore, not allowed to refer to your peers' work except those in your team or use any materials from books/online directly for submission. Students found to be dishonest will be subjected to disciplinary actions (**see *Academic Impropriety***). |

<table>
<tr><td>WHAT YOU SHOULD HAND IN</td><td><ul><li>Final deliverable: Software Design Document (SDD)</li><li>You are required to submit a soft copy of the SDD to your tutor (in pdf format). For this assignment, please use the provided SDD template, which comes together with the Cover Sheet and the Plagiarism Statement.</li></ul>Note :<ul><li>When submitting the coursework, you have committed that your work is original and complies with the rules and regulations on the assignment (refer to academic Impropriety)</li><li>Please refer to the Submission Deadline section for the assignment deadlines.</li></ul></td></tr>
<tr><td>ASSESSMENT CRITERIA</td><td><ul><li>Please refer to the Assignment Rubrics.</li><li>PASSING MARK: The passing mark for this assignment is 50 marks.</li></ul></td></tr>
<tr><td>ACADEMIC INTEGRITY AND PLAGIARISM</td><td><ul><li>Students may be required to redo the entire assignment if the contents of any part do not comply with the assignment's requirements as requested by the respective lecturer/tutor.</li><li>All students must comply with TAR UMT's plagiarism policy. Any attempt to cheat, plagiarise, collude, or otherwise gain an unfair advantage in the assessment will result in a penalty.</li><li>IMPORTANT: Students found to be dishonest are liable to disciplinary action.</li></ul></td></tr>
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## APPENDIX A :

## ASSIGNMENT DESCRIPTION

### A. Instructions

i. For this assignment, you are provided with seven (7) case studies covering a variety of software development problem domains. Each case study includes a description, key functionalities, and targeted end-users (**Attachment B**), as well as a Use Case Model (**Attachment C**).

ii. Each group must **choose a case study,** and **each member must be responsible for one (1) module** and the relevant use case description.

iii. The group then must construct a design model for the chosen case study, and all outcomes will be compiled in a mini Software Design Document (SDD). **Please use the SDD template** provided.

### B. Expected Outcome

The outcomes produced by each team will be grouped into two (2) categories as follows:

| CATEGORY | EXPECTED OUTCOME | DESCRIPTION | REFERENCE |
|---|---|---|---|
| Group outcome | One (1) architecture diagram for the case study | • The team must **identify two (2) key Quality Objectives** relevant to the case study.<br>• Then, using the existing architectural patterns, the team must **construct an architecture supporting the identified Quality Objectives**.<br>• The team must also **justify** their final architecture. | Chap. 1, 5 and 8.<br><br>External Website |
| | A set of mockup UI using Figma | • **Each member** must **develop at least two (2) mockup pages** for their module.<br>• When combined, all the **mockup pages must be consistent and cohesive, representing pages from a single system**. | Chap. 7 |
| Individual outcome | A set of activity diagrams | • **Each member must construct a set of activity diagrams for their module**.<br>• One (1) activity diagram for each use case description (eg, module A with 3 use case descriptions = 3 activity diagrams).<br>• **Identify all possible functions or process flows**. | Chap. 9 |
| | A class diagram using a design pattern | • **Each member must construct one (1) class diagram for their module.**<br>• The class diagram **must apply one (1) design pattern** | Chap. 6 & Refactoring Guru Website |

## C. Proposed Activity Timeline / Implementation Steps

| STEP | WEEK | ACTIVITY | NOTE |
|---|---|---|---|
| Step 1 | Week 2 | Discuss the case study with the team members and tutor.<br><br>Assigned the module to the team members | You may explore and search for further information to supplement the information provided in Attachment B and Attachment C |
| Step 2 | Week 2–3 | **Identify the Quality Objectives / Quality Attributes** appropriate for the system<br><br>Prepare **SDD Template : Chapter 2.2.1** | What are the two most important Quality Objectives/Quality Attributes for the system? |
| Step 3 | Week 3-4 | **Identify and analyse three (3) architectural patterns** that can support the identified Quality Attributes.<br><br>**Then select and justify one (1) MOST SUITABLE** architectural pattern for the system. Get the Tutor's feedback.<br><br>Prepare **SDD Template : Chapter 2.2.2 - 2.2.3** | The architectural patterns are not limited to patterns discussed in class notes. You can refer to other references, including the internet.<br><br>**This section is very critical. Please pay extra attention.** |
| Step 4 | Week 3-4 | **Construct activity diagrams** based on the use case descriptions in the assigned module.<br><br>**1 activity diagram for 1 use case description**<br><br>Prepare **SDD Template : Chapter 3.x.1** | If the use case description does not provide enough information, you may add new process flows or functionalities. |
| Step 5 | Week 4-6 | **Construct a class diagram** for your module. The class diagram must **implement one (1) design pattern**<br><br>Prepare **SDD Template : Chapter 3.x.2** | Refer to Refactoring Guru website.<br><br>You need to self-study 21 design patterns on the website and choose 1 for your class diagram |
| Step 6 | Week 5-6 | **Each member** must **construct a minimum of 2 mockup pages** for their module. All pages will be compiled as one.<br><br>Prepare **SDD Template : Chapter 4** | All members **must use Figma.** |
| Step 7 | Week 6 | Compile and finalise the software design document | **Submit only one (1) copy (.pdf)**<br>**Deadline: Friday, Week 6, 11.59pm** |

## D. SDD Template

i. For this project, you are provided with the following documents :
   - Template for the Software Design Document (SDD)

ii. The document is available in Google Classroom

## CASE STUDIES AND USE CASE MODEL

| ATTACHMENT | CASE STUDY TITLE | PROBLEM DESCRIPTION |
|---|---|---|
| **B1 & C1** | Mental Health Support App | Many individuals struggle with stress, anxiety, and depression, but lack access to professional help. A digital platform is needed to provide self-help tools, guided therapy, and community support. |
| **B2 & C2** | Assistive Learning Platform for People with Disabilities | Traditional e-learning platforms often lack accessibility features for people with visual, hearing, or cognitive impairments. A specialised platform is needed to improve their learning experience. |
| **B3 & C3** | Digital Preservation of Indigenous Languages | Many indigenous languages are at risk of extinction due to a lack of formal documentation and digital resources. A platform is needed to support language learning and preservation. |
| **B4 & C4** | AI-Powered Fake News Detector | Misinformation spreads rapidly online, misleading users and causing harm. A browser extension or mobile app is needed to verify news credibility. |
| **B5 & C5** | Digital Transformation for Local Artisans & Craftspeople | Many traditional artisans struggle to sell their products online due to a lack of digital skills and exposure. A platform is needed to bridge this gap. |
| **B6 & C6** | AI-Powered Personalised Career Guidance | Many students and professionals struggle to choose the right career path due to a lack of personalised insights. A career recommendation system is needed |
| **B7 & C7** | Gamified Rehabilitation for Stroke Patients | Stroke survivors need engaging rehabilitation exercises to improve motor skills, but often find traditional therapy repetitive and discouraging. A gamified rehab app is required. |

## B1: MENTAL HEALTH SUPPORT APP

### 1.1 System Overview & Objectives

The Mental Health Support App provides a digital platform offering self-help tools, guided therapy, and community support for individuals dealing with stress, anxiety, and depression. The goal is to increase accessibility to mental health resources and professional assistance, particularly for users who cannot easily access therapy in person.

### 1.2 Primary Users & Roles

| | |
|---|---|
| **End Users (Individuals)** | Use self-help tools, track moods, interact in support forums, and access counselling resources |
| **Therapists** | Provide online sessions, review user assessments, and manage appointments. |
| **System Administrator** | Manages content, user accounts, and data privacy controls |

### 1.3 System Modules

| | |
|---|---|
| **User Profile & Authentication Module** | Handles account creation, user preferences, and secure login |
| **Mood Tracking & Assessment Module** | Records daily moods, stress levels, and generates reports using simple analytics. |
| **Mindfulness & Therapy Module** | Provides guided exercises, relaxation sessions, and meditation resources. |
| **Peer Support Forum** | Anonymous message boards and chat spaces are moderated for safety |
| **AI Counsellor Chatbot** | Offers 24/7 initial counselling, screening questions, and recommends next steps. |
| **Appointment & Crisis Support Module** | Connects users with professional therapists or emergency helplines |

### 1.4 Detailed Process Flow

When a new user first downloads the app, they are prompted to create a secure account, complete a confidentiality agreement, and fill in an initial mental health self-assessment. The system evaluates their responses using an AI chatbot, which identifies potential indicators of stress or anxiety. Based on this initial analysis, the chatbot recommends a personalised wellness plan consisting of daily mindfulness activities, guided relaxation sessions, and journaling prompts. As users interact daily, they log their moods through graphical sliders and free-text reflections. The app stores these inputs, detects emotional trends, and generates weekly summaries viewable on the user dashboard.

If the AI detects a consistent negative pattern, such as rapidly declining mood scores or alarming keywords in journal entries, it triggers a background rule that prompts the user with a supportive message and displays emergency resources like crisis hotlines. At any point, users can open the peer support forum, where the system anonymises their identity and displays topic categories (e.g., anxiety, relationship stress). The forum module checks user safety filters before posting, ensuring content does not violate support community guidelines.

For professional assistance, users can access the therapy booking interface, view available therapists, and select a suitable time slot. Once confirmed, the system creates a virtual meeting link, sends notifications to both parties, and synchronises the appointment into the therapist's dashboard. Therapists can later view user assessment reports, make notes after sessions, and mark follow-up actions. Behind the scenes, the system continuously collects behavioural data to refine AI counselling recommendations, creating a closed-loop process that blends self-help, social interaction, and professional care.

### 1.5 Data Flow / Key Entities

| | |
|---|---|
| **User Data** | Profile, assessment results, and mood logs. |
| **Forum Data** | Posts, replies, reports |
| **Therapy Data** | Appointment records and chat history. |

### 1.6 System Behaviour & Automation

- AI chatbot analyses emotional tone in user messages.
- Automatic reminders for self-check-ins.
- Escalation alerts when mood trends decline significantly.

## B2: ASSISTIVE LEARNING PLATFORM FOR PEOPLE WITH DISABILITIES

### 2.1 System Overview & Objectives

This e-learning platform enhances accessibility for users with visual, hearing, or cognitive impairments. It delivers adaptive content formats, ensuring inclusive learning experiences through the use of assistive technologies.

### 2.2 Primary Users & Roles

| | |
|---|---|
| **Students with Disabilities** | Access course materials in their preferred assistive format. |
| **Instructors** | Upload learning materials, manage accessibility settings, and track learner progress. |
| **Accessibility Experts** | Evaluate content accessibility and suggest improvements |

### 2.3 System Modules

| | |
|---|---|
| **User Accessibility Configuration Module** | Allows users to customise font size, contrast, narration, or sign language overlay. |
| **Learning Content Management** | Hosts lessons, quizzes, videos, and transcripts |
| **Assistive Interface Engine** | Integrates text-to-speech, voice command navigation, and sign language videos |
| **Adaptive Learning & Gamification** | Adjusts quiz difficulty, provides badges, and encourages engagement |
| **Collaboration Tools** | Discussion boards and group workspaces with accessibility modes. |

### 2.4 Detailed Process Flow

Upon signing up, the student completes an accessibility profile questionnaire specifying their disability type, preferred interaction mode, and device capabilities. Based on this information, the system automatically configures the necessary interface to activate relevant features, such as text-to-speech for visually impaired users, sign language overlays for hearing-impaired learners, or simplified navigation for individuals with cognitive limitations. When the learner accesses a course, the Learning Management Module retrieves materials and dynamically reformats them according to the user's settings.

As the student progresses through lessons, the assistive interface continuously monitors interaction patterns. For example, if the system detects that the user frequently pauses or rewinds videos, it prompts the user to choose whether to reduce the playback speed or switch to transcript mode. During quizzes, the adaptive engine adjusts the difficulty of questions based on previous scores, ensuring both challenge and accessibility. Voice command integration enables visually impaired users to respond by speaking, which the system converts into text and validates.

Instructors log into their dashboards to upload multimedia lessons. The system automatically transcribes videos, adds subtitles, and checks compliance with accessibility guidelines before publishing. In collaborative group activities, the system assigns accessible interfaces for each participant, such as sign-language video for one and high-contrast visuals for another, ensuring synchronous collaboration. Accessibility experts periodically review system logs and generate reports identifying content accessibility issues or UI friction points. This end-to-end process ensures continuous feedback, adaptive delivery, and inclusivity for all learners.

### 2.5 Data Flow / Key Entities

| | |
|---|---|
| **User Profile Data** | Accessibility preferences, learning progress. |
| **Course Content** | Multimedia lessons and quiz metadata. |
| **Collaboration Data** | Forum and chat logs |

### 2.6 System Behaviour & Automation

- Automatic UI adaptation after each login.
- AI detects difficulty patterns and adjusts content delivery pace.

## B3: DIGITAL PRESERVATION OF INDIGENOUS LANGUAGES

### 3.1 System Overview & Objectives

A web and mobile platform designed to preserve and teach indigenous languages through interactive lessons, community contributions, and multimedia archives.

### 3.2 Primary Users & Roles

| | |
|---|---|
| **Language Learners** | Engage with interactive lessons and quizzes |
| **Community Contributors** | Add words, phrases, and cultural stories |
| **Linguists** | Validate entries and provide academic oversight |

### 3.3 System Modules

| | |
|---|---|
| **Language Learning Module** | Lessons, pronunciation guides, and speech recognition for evaluation |
| **Community Contribution Portal** | Enables crowdsourced dictionary entries and translations |
| **Cultural Archive** | Stores stories, songs, and videos |
| **Gamification & Progress Tracker** | Rewards users for learning and contributing. |
| **Offline Mode** | Allows learning without connectivity in remote areas |

### 3.4 Detailed Process Flow

When users first access the platform, they select an indigenous language and specify their proficiency level. The system then builds a customised learning path, starting with introductory vocabulary lessons supported by audio clips recorded by native speakers. Each lesson involves pronunciation practice, during which the speech recognition engine analyses the learner's microphone input to detect pronunciation deviations. The system provides real-time feedback, suggesting repeated attempts where errors are significant.

Community contributors can submit new words, idioms, or stories through a dedicated contribution form. Each submission includes text, translation, and optionally an audio or video recording. Once submitted, the entry enters a pending queue where linguists receive notifications. Linguists verify the linguistic accuracy, dialect category, and metadata before marking the entry as "Approved." The approved content is then automatically added to the public dictionary and made searchable for all users.

Simultaneously, the platform's cultural archive stores and categorises oral histories, traditional songs, and visual artefacts. When users engage with this content, the system tracks activity to provide recommendations such as "Explore related stories from the same region." Offline learning is supported by caching the user's lesson materials; synchronisation occurs once the device reconnects, updating both progress and community contributions. This cyclical workflow of learning, contributing, validating, and preserving will ensure that linguistic knowledge evolves in a collaborative and sustainable manner.

### 3.5 Data Flow / Key Entities

| | |
|---|---|
| **Language Data** | Word entries, grammar rules, audio clips |
| **User Data** | Learning history, badges |
| **Community Content** | Pending and approved submissions. |

### 3.6 System Behaviour & Automation

- AI flags duplicate or inaccurate contributions.
- Automatic content synchronisation for offline users.

## B4: AI-POWERED FAKE NEWS DETECTOR

### 4.1 System Overview & Objectives

A browser extension or mobile app that identifies misinformation by verifying news articles against credible databases and applying AI-driven credibility scoring.

### 4.2 Primary Users & Roles

| | |
|---|---|
| **General Users** | Check article authenticity |
| **Fact-Checkers** | Review flagged content |
| **Administrators** | Manage source databases |

### 4.3 System Modules

| | |
|---|---|
| **Article Analyzer** | Extracts text, title, and source metadata |
| **AI Credibility Engine** | Uses NLP and ML models to determine reliability |
| **Fact-Check Database** | Stores verified claims and sources |
| **Community Reporting Interface** | Allows users to flag suspicious posts |
| **Media Literacy Module** | Educates users on misinformation detection |

### 4.4 Detailed Process Flow

When a user installs the browser extension or mobile app, it integrates with the user's browsing environment. As the user reads an article, the system's analyser automatically extracts metadata such as the domain name, publication date, author, and body text. This data is sent to the AI Credibility Engine, which compares the article's claims against the Fact-Check Database, which aggregates verified statements from authoritative sources such as Reuters or Snopes. The AI model assigns a credibility score and returns visual feedback within seconds, for example, displaying a green badge for trustworthy articles, a yellow badge for uncertain content, and a red badge for suspicious content.

If the user clicks the badge, a detailed credibility report appears showing matched claims, sources, and similarity confidence levels. Users can also manually report content they believe is misleading. These reports trigger a workflow in which fact-checkers receive alerts, review the flagged material, cross-reference external databases, and either confirm or dispute the AI's decision. Once validated, the report feeds back into the model's training data, enabling iterative learning.

The system's community dashboard aggregates trends in flagged content, allowing administrators to track misinformation clusters or coordinated campaigns. For educational purposes, the system provides an interactive learning centre that simulates comparisons between fake and real articles, helping users understand detection indicators. Through continuous monitoring, feedback loops, and AI retraining, the process achieves both automated detection and human-verified accuracy.

### 4.5 Data Flow / Key Entities

| | |
|---|---|
| **News Data** | URLs, text content, metadata |
| **Credibility Results** | Score, explanations |
| **User Reports** | Flag history and comments |

### 4.6 System Behaviour & Automation

- Automated scanning of new URLs shared by users.
- Continuous model improvement based on verified results.

## B5: DIGITAL TRANSFORMATION FOR LOCAL ARTISANS & CRAFTSPEOPLE

### 5.1 System Overview & Objectives

A digital commerce platform enabling local artisans to showcase and sell handmade products online, with marketing, training, and multilingual support.

### 5.2 Primary Users & Roles

| | |
|---|---|
| **Artisans / Sellers** | Manage storefronts and upload products |
| **Buyers** | Browse and purchase crafts |
| **Trainers / Admins** | Conduct virtual workshops and provide technical help |

### 5.3 System Modules

| | |
|---|---|
| **Storefront Management** | Product upload, pricing, and inventory tracking |
| **Payment Gateway Integration** | Supports secure transactions with multiple currencies |
| **AI Recommendation Engine** | Suggests products to buyers and marketing strategies to sellers |
| **Social Media Integration** | Auto-posts listings and promotes artisan stories |
| **Learning & Workshop Portal** | Offers business tutorials and webinars |

### 5.4 Detailed Process Flow

An artisan begins by registering on the platform, entering personal details, workshop location, and payment credentials. The system validates these details and generates a personalised storefront template. Using the dashboard, the artisan uploads product photos, sets prices, and writes descriptions. The platform's AI engine automatically tags items with relevant categories and recommends optimal price ranges based on market trends.

When the store goes live, buyers can browse through categories, search for artisans by location or product type, and add items to their cart. The system displays AI-based product recommendations on each page to increase visibility. Once an order is placed, the payment gateway securely processes the transaction and sends confirmation messages to both the buyer and the seller. Simultaneously, stock quantities are automatically updated.

For artisans unfamiliar with digital marketing, the system suggests promotional actions, such as cross-posting products to Instagram or scheduling participation in virtual fairs. The platform hosts workshops that artisans can join via live video, where trainers teach business and packaging skills. After each workshop, a feedback form is sent, and completion is recorded in the artisan's learning record. The entire cycle, from onboarding to sales analytics and capacity building, is continuously supported by an AI recommendation layer that monitors performance and offers data-driven advice.

### 5.5 Data Flow / Key Entities

| | |
|---|---|
| **Product Data** | Descriptions, prices, and inventory |
| **User Data** | Seller/buyer profiles, transaction logs. |
| **Training Data** | Course materials and attendance |

### 5.6 System Behaviour & Automation

- AI-driven marketing prompts and stock notifications.
- Fraud detection and payment reconciliation automation.

## B6: AI-POWERED PERSONALIZED CAREER GUIDANCE

### 6.1 System Overview & Objectives

A web platform offering AI-driven career recommendations based on users' skills, interests, and experience, integrating with job and mentorship opportunities.

### 6.2 Primary Users & Roles

| | |
|---|---|
| **Students / Job Seekers** | Complete assessments and receive guidance |
| **Career Counsellors** | Validate AI results, offer feedback |
| **Recruiters / Mentors** | Post internships, connect with users |

### 6.3 System Modules

| | |
|---|---|
| **Skill and Interest Assessment Engine** | Uses questionnaires and portfolio data to analyse user potential |
| **AI Career Matching Module** | Matches users to career paths, courses, and roles |
| **Learning Recommendation Engine** | Suggests online courses to close skill gaps |
| **Mentorship and Job Portal** | Connects users with professionals or employers |
| **Progress Dashboard** | Tracks learning achievements and career growth |

### 6.4 Detailed Process Flow

Upon registration, the user fills out a detailed profile that captures their educational background, current skills, interests, and personality indicators. The Skill Assessment Engine presents interactive questions, scenario-based tasks, or portfolio uploads to gauge technical and soft skills. Once submitted, the AI Career Matching Module analyses results against a database of career archetypes and labour market data to produce a ranked recommendation list.

The user views these results through a visual dashboard, exploring each career suggestion with details on salary trends, demand forecasts, and required competencies. If the AI detects gaps between the user's current and desired skill sets, the Learning Recommendation Engine proposes specific online courses, certifications, or university programs. Users can bookmark recommendations and track their progress in their career development roadmap.

Career counsellors can log in to review AI recommendations for their assigned users, make manual adjustments, and conduct chat or video consultations. Mentors and recruiters, meanwhile, have access to filtered lists of candidates based on matching criteria, allowing them to invite users to mentorship sessions or job interviews. As users complete courses or internships, the system automatically updates their profiles, recalculates readiness scores, and sends periodic insights summarising progress and new career matches.

### 6.5 Data Flow / Key Entities

| | |
|---|---|
| **User Profile** | Skills, interests, test results |
| **Career Data** | Job roles, required competencies |
| **Learning Resources** | Courses, mentors, and progress records. |

### 6.6 System Behaviour & Automation

- Adaptive algorithm refines career suggestions with user feedback.
- Notification system prompts users to update their profiles periodically.

## B7: GAMIFIED REHABILITATION FOR STROKE PATIENTS

### 7.1 System Overview & Objectives

A rehabilitation app that uses gamification and motion tracking to help stroke patients recover motor skills through engaging, measurable exercises.

### 7.2 Primary Users & Roles

| | |
|---|---|
| **Stroke Patients** | Perform daily therapy exercises |
| **Physiotherapists** | Assign tasks and monitor recovery |
| **Caregivers** | Support patient progress tracking |

### 7.3 System Modules

| | |
|---|---|
| **Motion Tracking & Exercise Module** | Uses smartphone camera or wearables for movement capture |
| **AI Progress Tracker** | Monitors exercise performance and improvement |
| **Game Engine & Reward System** | Converts exercises into fun, level-based games |
| **Therapist Dashboard** | Provides real-time feedback and personalised plans |
| **Remote Therapy Module** | Enables live video sessions between the therapist and the patient |

### 7.4 Detailed Process Flow

When a physiotherapist registers a new patient, the system records the patient's baseline mobility scores and creates an exercise plan tailored to the patient’s affected limb or movement type. The patient accesses the app on a smartphone or tablet and begins sessions using on-screen guidance. As they move, the device’s camera or connected wearable sensors capture motion data, which the AI engine analyses in real time to measure precision, range, and speed.

The app visualises each movement as a game, for instance, lifting an arm to collect virtual stars or balance exercises represented as stability challenges. Every successful movement earns points and triggers encouraging feedback. When the system detects incorrect movement patterns or excessive fatigue, it pauses the session and displays rest prompts to encourage a break.

All exercise data is synchronised to the physiotherapist’s dashboard, where progress graphs show accuracy improvements over time. Therapists can remotely adjust exercise intensity or assign new routines. Caregivers also receive summarised daily reports to help monitor adherence. If a patient repeatedly skips sessions, automated reminders and motivational notifications are triggered. Through continuous sensing, gamification, and remote oversight, the rehabilitation process becomes both personalised and measurable.

### 7.5 Data Flow / Key Entities

| | |
|---|---|
| **Patient Data** | Medical profile, exercise logs |
| **Therapy Data** | Assigned routines, progress metrics |
| **Gamification Data** | Points, badges, rankings |

### 7.6 System Behaviour & Automation

- AI adapts exercise difficulty based on real-time performance.
- Automatic reminders for missed sessions.

## C1: USE CASE MODEL FOR MENTAL HEALTH SUPPORT APP

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| **MHS-UC-01** | Register / Authenticate User | End User, System Admin | Create a secure account, gather baseline consent and profile data |
| **MHS-UC-02** | Complete Mental Health Assessment | End User, AI Chatbot | Collect initial screening data to personalise the care plan |
| **MHS-UC-03** | Daily Mood Logging & Journal | End User | Capture mood entries and free-text journals for trend analysis |
| **MHS-UC-04** | AI Counselling Session | End User, AI Chatbot | Provide immediate supportive conversation and triage |
| **MHS-UC-05** | Book Therapy Appointment | End User, Therapist, Scheduler | Schedule virtual/in-person therapy sessions |
| **MHS-UC-06** | Participate in Anonymous Forum | End User, Moderator | Post and interact in anonymised peer support spaces |
| **MHS-UC-07** | Crisis Escalation / Emergency Support | End User, System | Detect and respond to high-risk situations |
| **MHS-UC-08** | Therapist Review & Session Notes | Therapist | Review client history and add session notes |

| **Use Case ID** | **MHS-UC-01** |
|---|---|
| **Preconditions** | User has the app or web access; system is accessible; privacy policy is available. |
| **Trigger** | User chooses to register or log in. |
| **Main Flow** | 1. User opens app and selects "Register".<br>2. System displays privacy/consent and collects consent.<br>3. User provides email/phone, password and basic demographics (age bracket, location).<br>4. System validates input, enforces password rules, and runs optional 2FA ( SMS/email).<br>5. System creates account, stores encrypted credentials, and presents initial onboarding wizard to set notification preferences and emergency contact.<br>6. System logs creation and notifies admin if required (e.g., corporate deployment). |
| **Alternate Flows** | A1: User chooses social login — system authenticates via OAuth and maps profile.<br>A2: 2FA fails — the user can request a resend or an alternate verification method. |
| **Postconditions** | Account created; user session started; minimal profile stored; consent recorded. |
| **Special Requirements** | PII must be encrypted and comply with GDPR/PDPA; an optional anonymous mode is available for forums. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **MHS-UC-02** |
|---|---|
| **Preconditions** | User authenticated; baseline consent exists. |
| **Trigger** | User initiates assessment (onboarding or periodic reassessment) or chatbot recommends it. |
| **Main Flow** | 1. The user is guided through a structured assessment (multi-page questionnaire).<br>2. Each response is validated as entered; the AI chatbot monitors responses in real time for red flags (self-harm, suicidal ideation). |

| | |
|---|---|
| | 3. On completion, the AI computes scores (anxiety, depression indices), generates a summary, and recommends a personalized care path (self-help modules, therapist booking, or immediate escalation) which is shown to the user with rationale.<br>4. Results are stored in the user's secure record and flagged for therapist review if thresholds are crossed. |
| **Alternate Flows** | AF1: User abandons mid-assessment — system saves partial responses and prompts to continue later.<br>AF2: Red-flag triggers immediate crisis path. |
| **Postconditions** | Assessment data stored; care recommendations generated; flags/alerts created if necessary. |
| **Special Requirements** | Scoring algorithm must be auditable; red-flag response time < 5 seconds; secure storage. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **MHS-UC-03** |
|---|---|
| **Preconditions** | User authenticated and has access to dashboard. |
| **Trigger** | User opens the mood tracker or receives scheduled reminder. |
| **Main Flow** | 1. The user selects mood via slider or emoji, optionally tags activities, and writes a short journal entry.<br>2. The system timestamps entries, runs NLP sentiment analysis on free text (to surface keywords), and stores structured mood history.<br>3. If patterns show deterioration, the system surfaces suggested self-help modules or prompts to contact a therapist.<br>4. Weekly summary charts are computed and made available in the dashboard. |
| **Alternate Flows** | AF1: Offline entry — data cached and synced when online.<br>AF2: User enables sharing of select mood entries with therapist — system requests permission before transmitting. |
| **Postconditions** | Mood entry recorded; analytics updated; triggers raised if thresholds met. |
| **Special Requirements** | Sentiment analysis must not store raw text beyond the retention policy unless permitted. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **MHS-UC-04** |
|---|---|
| **Preconditions** | User authenticated; AI service operational; user consented to automated counselling. |
| **Trigger** | User initiates chat or system auto-starts when concerning patterns detected. |
| **Main Flow** | 1. The user starts a chat; the chatbot greets and asks screening questions to understand the issue.<br>2. The AI analyzes user language, suggests coping techniques (breathing exercises, grounding), and provides immediate resources.<br>3. If user expresses severe risk, the AI follows emergency escalation rules: it provides crisis hotline info, offers to connect to a live therapist, and requests permission to alert emergency contact if immediate danger is detected.<br>4. A transcript (subject to privacy settings) is stored for therapist review if the user consents. |

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| **Alternate Flows** | AF1: User requests human therapist connection — schedule flow initiated.<br>AF2: User disables AI — session ends with a resource list. |
| **Postconditions** | Chat summary stored per user settings; escalation actions completed if needed. |
| **Special Requirements** | AI responses must include confidence scores; safety filters to avoid harmful advice. |
| **Frequency** | Medium |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **MHS-UC-05** |
| **Preconditions** | Therapist profiles available; calendar integration configured; payment methods if needed. |
| **Trigger** | User chooses "Book Appointment" from recommendations or manual search. |
| **Main Flow** | 1. User searches therapists by specialization, availability, fee, and language.<br>2. The system shows available slots, and the user selects a slot and confirms payment (if applicable).<br>3. System books the slot, creates a virtual meeting link, sends confirmations to both parties, and stores the appointment in the therapist dashboard and user calendar.<br>4. Reminders are sent per user preferences.<br>5. Post-session, the therapist can add notes and follow-up actions. |
| **Alternate Flows** | AF1: Therapist cancels — system proposes alternate slots and notifies the user; refund processed if paid.<br>AF2: Payment failure — booking held for a grace period or canceled. |
| **Postconditions** | Appointment created or rescheduled; notifications sent. |
| **Special Requirements** | Calendar sync (iCal/Google), secure meeting link generation, HIPAA/PDPA-secure communication. |
| **Frequency** | Medium |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **MHS-UC-06** |
| **Preconditions** | Forum open; content policy and moderation rules defined. |
| **Trigger** | User navigates to forum and chooses to post or reply. |
| **Main Flow** | 1. The user creates a post or reply with optional tags.<br>2. System anonymizes identity (unless user opts out), runs content safety filters (detecting self-harm, hate speech), and queues posts for moderation if necessary.<br>3. Approved content is displayed with community voting and reply threads.<br>4. Moderators can flag and remove content; repeat offenders are warned or restricted.<br>5. The system logs moderation actions and provides summary metrics to admins. |
| **Alternate Flows** | AF1: Post flagged for immediate review — moderator receives notification.<br>AF2: User violates rules — system issues warning, temporary ban or escalate to admin. |
| **Postconditions** | Post published or rejected; moderation logs updated. |
| **Special Requirements** | Anonymous posting must not be reversible except under legal process; robust moderation tools. |
| **Frequency** | Medium |
| **Priority** | Medium |

| **Use Case ID** | **MHS-UC-07** |
|---|---|
| **Preconditions** | Emergency contact details provided (optional); crisis hotline integration configured. |
| **Trigger** | Red-flag from assessment or conversation (keywords indicating harm) or extremely low mood trends over multiple days. |
| **Main Flow** | 1. The system detects crisis signals and immediately interrupts normal workflows to show high-priority alerts and emergency actions.<br>2. It displays crisis resources, prompts the user to call a hotline, offers to initiate c ontact with a live therapist, and, with explicit consent, notifies emergency contacts or local emergency services.<br>3. All system actions are logged with timestamps.<br>4. Therapists assigned to the user receive an urgent notification for follow-up. |
| **Alternate Flows** | AF1: User declines escalation — system offers safety plan and schedules a re-check.<br>AF2: No emergency contact provided — the system focuses on hotlines and prompts users to provide a contact. |
| **Postconditions** | Emergency resources presented; notifications sent; incident logged. |
| **Special Requirements** | Immediate response SLA; legal/regulatory compliance for escalation. |
| **Frequency** | Low |
| **Priority** | Critical |

| **Use Case ID** | **MHS-UC-08** |
|---|---|
| **Preconditions** | Therapist assigned and authenticated; user consent to share assessment data given. |
| **Trigger** | Therapist opens the client profile or completes a session. |
| **Main Flow** | 1. The therapist accesses the user's assessment history, mood logs, and AI summaries.<br>2. They add session notes, mark treatment goals, and schedule follow-ups.<br>3. Therapists can flag unusual behaviour and recommend changes to the care plan; such recommendations propagate into the user's dashboard and trigger notifications.<br>4. The system securely stores notes and allows for export of clinical records if permitted. |
| **Alternate Flows** | AF1: Therapist adds sensitive notes — stored with higher access control or redaction available. |
| **Postconditions** | Clinical notes saved; follow-ups scheduled; care plan updated. |
| **Special Requirements** | Audit trail for notes; role-based access control; retention policy. |
| **Frequency** | Medium |
| **Priority** | High |

## C2: USE CASE MODEL FOR ASSISTIVE LEARNING PLATFORM FOR PEOPLE WITH DISABILITIES

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| ALP-UC-01 | Create / Configure Accessibility Profile | Student, System | Capture disability type and preferred accessibility settings |
| ALP-UC-02 | Upload & Auto-Format Course Material | Instructor, System | Upload content and auto-generate accessible formats |
| ALP-UC-03 | Access Lesson with Assistive Interface | Student, Assistive Engine | Deliver lessons adapted to student preferences |
| ALP-UC-04 | Voice Command Navigation | Student | Navigate the platform using voice commands |
| ALP-UC-05 | Adaptive Quiz / Assessment | Student, Adaptive Engine | Dynamically adjust difficulty and format to learner needs |
| ALP-UC-06 | Instructor Review & Accessibility QA | Instructor, Accessibility Expert | Validate and fix content accessibility issues |
| ALP-UC-07 | Collaborative Activity with Accessibility | Student, Instructor | Enable group work with mixed-accessibility modes |

| **Use Case ID** | **ALP-UC-01** |
|---|---|
| **Preconditions** | New or returning student with account. |
| **Trigger** | Student signs in and chooses to set accessibility preferences. |
| **Main Flow** | 1. Student answers a guided questionnaire covering vision, hearing, cognitive needs, and device constraints.<br>2. The system recommends presets (e.g., high contrast, TTS enabled) and allows fine-grained settings changes (font size, spacing, sign-language overlay).<br>3. The chosen settings are stored in the profile and immediately applied to the UI and content rendering engine. The system offers a quick preview for verification and allows the creation of multiple profiles (e.g., "Mobile", "Tablet"). |
| **Alternate Flows** | AF1: Student skips configuration — system uses default profile with recommendations to configure later. |
| **Postconditions** | Accessibility settings saved and active. |
| **Special Requirements** | Settings should be exportable/importable, and should persist across devices. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **ALP-UC-02** |
|---|---|
| **Preconditions** | Instructor authenticated; file upload enabled. |
| **Trigger** | Instructor uploads content (video, PDF, slides). |
| **Main Flow** | 1. The system ingests the content, auto-generates transcripts for audio/video, generates alt-text for images using metadata and prompts, and produces simplified HTML or ePub versions for better readability.<br>2. It runs an accessibility checker and returns a compliance report to the instructor with suggested edits. |

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| | 3. Once instructor approves, the platform publishes accessible versions and links them to lesson modules. |
| **Alternate Flows** | AF1: Auto-transcription confidence low — system requests human review or suggests engaging an accessibility expert. |
| **Postconditions** | Accessible materials published; compliance report stored. |
| **Special Requirements** | Transcriptions must support multiple languages; accuracy threshold documented. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **ALP-UC-03** |
|---|---|
| **Preconditions** | Student profile set; lesson published. |
| **Trigger** | Student clicks on a lesson. |
| **Main Flow** | 1. The system retrieves the lesson and renders it per profile: text rendered in large fonts, images with alt-text, video with sign language overlay or captions. 2. The TTS engine reads content aloud; the UI highlights focus items for keyboard navigation.<br>3. Interaction controls are adapted (larger touch targets, simplified menus).<br>4. The system logs interaction metrics (time on content, skips) to inform future adaptations. |
| **Alternate Flows** | AF1: Network constraints — fallback to cached low-bandwidth mode. |
| **Postconditions** | Lesson presented in accessible form; interaction logged. |
| **Special Requirements** | Low-latency switching of accessibility modes; consistent UI. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **ALP-UC-04** |
|---|---|
| **Preconditions** | Microphone access granted; voice engine initialized. |
| **Trigger** | Student activates voice mode or says a wake word. |
| **Main Flow** | 1. The student issues natural language navigation commands (e.g., "Open lesson 3", "Next slide", "Start quiz").<br>2. The system performs intent classification, executes the command, and confirms via audio/visual feedback.<br>3. Ambiguous commands provoke a clarifying question.<br>4. All voice commands and confirmations are logged. |
| **Alternate Flows** | AF1: Speech recognition fails — the system displays on-screen command options. |
| **Postconditions** | Navigation completed; accessibility logs updated. |
| **Special Requirements** | Voice recognition must support local accents; offline capabilities beneficial. |
| **Frequency** | Medium |
| **Priority** | Medium |

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| **Use Case ID** | **ALP-UC-05** |
| **Use Case Name** | Adaptive Quiz / Assessment |
| **Trigger** | Student starts an assessment. |
| **Main Flow** | 1. The system selects an initial question based on profile.<br>2. If the student answers correctly, the engine increases difficulty; if incorrect, it adapts to simpler phrasing or alternative modalities (audio instead of visual).<br>3. Time limits and help hints are offered per accessibility settings. Results are scored and stored; remediation suggestions appear automatically. |
| **Alternate Flows** | AF1: Student requests extended time — system checks policy and applies accommodation. |
| **Postconditions** | Scores stored; personalized remedial path recommended. |
| **Special Requirements** | Must support policy-based accommodations and audit trails. |
| **Frequency** | Medium |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **ALP-UC-06** |
| **Preconditions** | Uploaded content available. |
| **Trigger** | Instructor requests or receives an accessibility report. |
| **Main Flow** | 1. Accessibility expert or automated engine reviews flagged items; the instructor edits or replaces assets as recommended.<br>2. The system re-runs checks until compliance thresholds are met.<br>3. Final approval publishes content with a compliance badge. |
| **Alternate Flows** | AF1: Legal/regulatory non-compliance requires escalation to admin. |
| **Postconditions** | Lesson marked accessible; record of changes stored. |
| **Special Requirements** | Versioning of materials; audit logs for compliance. |
| **Frequency** | Low |
| **Priority** | Medium |

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| **Use Case ID** | **ALP-UC-07** |
| **Preconditions** | Group activity scheduled and participants have varied accessibility profiles. |
| **Trigger** | Group activity begins. |
| **Main Flow** | 1. The system assigns roles and creates an activity workspace where each participant's interface is adapted. For example, one member receives sign-language video for a presentation while another receives text annotations.<br>2. Shared artifacts (whiteboard, documents) are presented in accessible formats.<br>3. Real-time communication uses captions and audio cues as needed.<br>3. The instructor monitors participation and can adjust assignments if a participant is having trouble. |
| **Alternate Flows** | AF1: Participant loses connection — system records contributions and allows rejoin. |
| **Postconditions** | Activity completed; accessibility participation metrics recorded. |
| **Special Requirements** | Real-time captioning and low latency; cross-device consistency. |
| **Frequency** | Medium |
| **Priority** | Medium |

## C3: USE CASE MODEL FOR DIGITAL PRESERVATION OF INDIGENOUS LANGUAGES

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| DPL-UC-01 | Register & Choose Language Path | Learner, System | Set language and proficiency and create learning path |
| DPL-UC-02 | Take Pronunciation Exercise & Feedback | Learner, Speech Engine | Practice and receive pronunciation evaluation |
| DPL-UC-03 | Contribute Word / Story | Community Contributor | Submit language items to the repository |
| DPL-UC-04 | Linguist Review & Approve Content | Linguist | Validate and approve community contributions |
| DPL-UC-05 | Access Cultural Archive (Offline Sync) | Learner | Download content for offline use in remote areas |
| DPL-UC-06 | Gamified Learning & Rewards | Learner, System | Encourage participation and contributions |
| DPL-UC-07 | Search & Translate Dictionary | Learner, Contributor | Lookup terms and request translations |

| **Use Case ID** | **DPL-UC-01** |
|---|---|
| **Preconditions** | Platform accessible; language options available. |
| **Trigger** | New learner signs up or selects language. |
| **Main Flow** | 1. Learner registers (or logs in), selects target indigenous language and indicates proficiency level and learning goals.<br>2. The system generates a recommended learning path (vocabulary sets, cultural modules) and offers an introductory lesson with native-speaker audio.<br>3. The learner can choose to follow the suggested path or customise modules; progress tracking begins immediately. |
| **Alternate Flows** | AF1: Learner requests community mentor — system suggests local contributors or linguists. |
| **Postconditions** | Learning path created; initial lesson unlocked. |
| **Special Requirements** | Multi-language UI; offline-first packaging for remote deployments. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **DPL-UC-02** |
|---|---|
| **Preconditions** | Microphone permission granted; exercise available. |
| **Trigger** | Learner opens pronunciation module. |
| **Main Flow** | 1. The system plays native-speaker audio and displays text.<br>2. The learner records attempts; the speech engine analyses phonemes against expected patterns and returns a score plus targeted tips (which syllables were off).<br>3. The learner is offered repeat attempts and receives badges for milestone improvement; data is stored for longitudinal tracking. |
| **Alternate Flows** | AF1: Low audio quality — system prompts for re-record or suggests moving to a quieter environment. |
| **Postconditions** | Attempt logged; feedback stored; progress updated. |

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| **Special Requirements** | Support for low-bandwidth audio; tolerant algorithms for dialect variants. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **DPL-UC-03** |
|---|---|
| **Preconditions** | Contributor authenticated or contribution allowed anonymously per community policy. |
| **Trigger** | Contributor chooses "Add Entry" in the dictionary or archive. |
| **Main Flow** | 1 Contributor fills a form with the word/phrase, audio/video, contextual sentence, and metadata (dialect, region).<br>2. The system captures the submission and assigns a temporary ID and status "Pending Review."<br>3. A notification is sent to linguists and local community curators for vetting.<br>4. Contributors earn reputation points for submissions. |
| **Alternate Flows** | AF1: Contributor requests immediate publishing for urgent cultural items — escalation to curator. |
| **Postconditions** | Entry queued for review; contributor credited. |
| **Special Requirements** | Metadata schema; consent for audio/video use. |
| **Frequency** | Medium |
| **Priority** | Medium |

| **Use Case ID** | **DPL-UC-04** |
|---|---|
| **Preconditions** | Pending contributions in queue. |
| **Trigger** | Linguist receives notification or opens review dashboard. |
| **Main Flow** | 1. Linguist examines submission, verifies orthography, checks audio accuracy, adds notes or corrections, and approves or rejects.<br>2. If approved, the entry is published to the public dictionary and linked to associated lessons. If rejected, feedback is sent to contributor for revision.<br>3. The system records reviewer identity and timestamp for provenance. |
| **Alternate Flows** | AF1: Disputed item — the system flags for community discussion and consensus-based resolution. |
| **Postconditions** | Item moved to approved or rejected state; provenance logged. |
| **Special Requirements** | Version history and provenance metadata required. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **DPL-UC-05** |
|---|---|
| **Preconditions** | Learner profile exists; device storage available. |
| **Trigger** | Learner selects content for offline download or is scheduled for sync. |
| **Main Flow** | 1. Learner selects items to cache (stories, audio, lessons).<br>2. The system packages selected content with metadata, compresses files for offline use, and starts download.<br>3. On reconnection, the system synchronizes annotations and contributions made offline, handles merge conflicts if the same item was updated online, and updates local progress. |

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| **Alternate Flows** | AF1: Insufficient storage — system prompts to deselect items or connect external storage. |
| **Postconditions** | Content available offline; sync logs updated when reconnected. |
| **Special Requirements** | Conflict resolution policies; low-bandwidth packaging. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **DPL-UC-06** |
|---|---|
| **Preconditions** | Learner enrolled and gamification enabled. |
| **Trigger** | Learner completes tasks or contributes content. |
| **Main Flow** | 1. System awards points/badges based on milestone events (first 50 words learned, first approved contribution).<br>2. Leaderboards and achievement pages update. Badges unlock content and community recognition.<br>3. System periodically runs events (e.g., "Story Month") to encourage contributions. |
| **Alternate Flows** | AF1: Learner wants privacy — can opt out of leaderboards while retaining badges privately. |
| **Postconditions** | Rewards recorded; progression updated. |
| **Special Requirements** | Mechanism to prevent gaming the system; opt-out privacy options. |
| **Frequency** | Medium |
| **Priority** | Medium |

| **Use Case ID** | **DPL-UC-07** |
|---|---|
| **Preconditions** | Dictionary populated. |
| **Trigger** | User enters search term. |
| **Main Flow** | 1. System performs morphological-aware search across entries, returns definitions, pronunciations, related items, and audio.<br>2. If the translation is not found, the user can request a community translation, generating a contributor task.<br>2. Search supports fuzzy matching and dialect filters. |
| **Alternate Flows** | AF1: No matches found — prompt to add a contribution. |
| **Postconditions** | Search results presented; requests logged if needed. |
| **Special Requirements** | Efficient indexing and offline-search capabilities. |
| **Frequency** | High |
| **Priority** | High |

## C4: USE CASE MODEL FOR AI-POWERED FAKE NEWS DETECTOR

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| FND-UC-01 | Install & Configure Detector | General User, System | Add extension/app and set preferences |
| FND-UC-02 | Analyze Article (Auto) | General User, System | Automatic scanning of opened article and scoring |
| FND-UC-03 | Request Detailed Credibility Report | General User | Get claim-level evidence and sources |
| FND-UC-04 | Submit User Report / Flag Content | User, Moderator, Fact-Checker | Flag suspected misinformation |
| FND-UC-05 | Fact-Checker Review & Verdict | Fact-Checker | Investigate flagged items and publish verdict |
| FND-UC-06 | Update Model & Source Database | Admin, System | Retrain models and curate sources |
| FND-UC-07 | Media Literacy Tutorial | User, System | Teach users to identify misinformation |

| **Use Case ID** | **FND-UC-01** |
|---|---|
| **Preconditions** | Browser or mobile OS supports extension/app. |
| **Trigger** | User installs extension or app. |
| **Main Flow** | 1. User installs the extension/app, grants permissions (page access, network calls).<br>2. The system walks the user through privacy options (share URLs for analysis, anonymous reporting).<br>3. The user configures sensitivity, preferred sources, and whether to auto-scan or scan on demand.<br>4. System validates settings and displays the status icon/toolbar. |
| **Alternate Flows** | AF1: User denies permissions — limited functionality provided with clear messaging. |
| **Postconditions** | Detector installed with configured preferences. |
| **Special Requirements** | Minimal performance impact; clear privacy disclosures. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **FND-UC-02** |
|---|---|
| **Preconditions** | Extension installed; network accessible; model services available. |
| **Trigger** | User opens an article URL or highlights text and invokes analysis. |
| **Main Flow** | 1. The extension extracts title, author, publish date, body text, and images.<br>2. The AI Credibility Engine checks claims against the fact-check database and external reputable sources, computes a credibility score, and renders an inline badge with an explanation summary.<br>3. The detailed report (if requested) includes matched claims, source citations, and confidence levels. |
| **Alternate Flows** | AF1: Paywalled article — system requests user to paste text.<br>AF2: Model unavailable — system queues the URL for later analysis and notifies the user. |

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| **Postconditions** | Credibility score displayed; logs updated. |
| **Special Requirements** | Fast response (< 3s ideally); transparent scoring explanations. |
| **Frequency** | High |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **FND-UC-03** |
| **Preconditions** | Article previously analyzed or live connection to service. |
| **Trigger** | User clicks “View Report.” |
| **Main Flow** | 1. System compiles evidence links, highlights matching claims, shows the fact-check database results and relevant source snippets, lists contradictions, and recommends trust level.<br>2. Users can download or share the report. |
| **Alternate Flows** | AF1: Sources behind paywalls — system summarises provenance and suggests alternative public references. |
| **Postconditions** | Report displayed and optionally saved/exported. |
| **Special Requirements** | Citation provenance; limit quoting to copyright allowances. |
| **Frequency** | Medium |
| **Priority** | Medium |

| | |
|---|---|
| **Use Case ID** | **FND-UC-04** |
| **Preconditions** | User authenticated (optional) and extension configured to allow reporting. |
| **Trigger** | User clicks "Flag" or fills a report form. |
| **Main Flow** | 1. User selects reason and adds comments and optionally attaches evidence.<br>2. The system logs the report, increases the article’s review priority, and notifies fact-checkers.<br>3. It provides the reporter with a tracking ID and optional follow-up when investigation completes. |
| **Alternate Flows** | AF1: Abuse of reporting — reports flagged for spam and reporter rate-limited. |
| **Postconditions** | Report logged; fact-check queue updated. |
| **Special Requirements** | Rate-limiting; anonymity options; moderation workflow. |
| **Frequency** | Medium |
| **Priority** | Medium |

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|---|---|
| **Use Case ID** | **FND-UC-05** |
| **Preconditions** | Report exists or items auto-flagged by the system. |
| **Trigger** | Fact-checker picks an item from the queue. |
| **Main Flow** | 1. Fact-checker examines claims, cross-references primary sources, documents findings, and issues a verdict (True/False/Misleading/Unverified) with linked references.<br>2. The verdict is published to the database and drives updates to the AI model training set.<br>3. The system also notifies the reporter (if requested) and updates the article badge. |

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| **Alternate Flows** | AF1: Inconclusive — publish "Unverified" with details and request community input. |
| **Postconditions** | Verdict stored; source citations recorded; model dataset updated. |
| **Special Requirements** | Verifiability, audit trail, and provenance metadata. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **FND-UC-06** |
|---|---|
| **Preconditions** | New fact-checks or curated sources available. |
| **Trigger** | Scheduled retraining or new validated data arrives. |
| **Main Flow** | 1. Admin curates authoritative sources and tags new fact-check entries.<br>2. System schedules model retraining, validates against hold-out sets, and upon successful validation deploys updated models.<br>3. A rollback plan exists if performance regresses. |
| **Alternate Flows** | AF1: Retraining fails validation — admin reviews data and adjusts parameters. |
| **Postconditions** | Model updated; database refreshed. |
| **Special Requirements** | Transparent model versioning, rollback, and monitoring. |
| **Frequency** | Periodic |
| **Priority** | High |

| **Use Case ID** | **FND-UC-07** |
|---|---|
| **Preconditions** | Tutorial module available. |
| **Trigger** | User opens the media literacy center. |
| **Main Flow** | 1. System presents interactive modules that show examples of manipulation, prompt the user to spot errors, and provide exercises.<br>2. Completion provides certification/badges and improves the user's detector preferences via optional quizzes. |
| **Alternate Flows** | AF1: User requests an instructor-led session — schedule flow triggered. |
| **Postconditions** | Tutorial completion recorded; user proficiency tracked. |
| **Special Requirements** | Engaging UX; accessible examples; evidence-based curriculum. |
| **Frequency** | Low |
| **Priority** | Low |

## C5: USE CASE MODEL FOR DIGITAL TRANSFORMATION FOR LOCAL ARTISANS & CRAFTSPEOPLE

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| DTA-UC-01 | Artisan Onboarding & Storefront Setup | Artisan, System | Help artisans create storefronts and configure payments |
| DTA-UC-02 | Upload Product Listing & Auto-Tag | Artisan, System | Publish product with automatic categorisation & optimisation |
| DTA-UC-03 | Buyer Search, Purchase & Payment | Buyer, System, Payment Gateway | Enable browsing, checkout, and secure payment |
| DTA-UC-04 | Order Fulfillment & Shipping | Artisan, Logistics, Buyer | Manage fulfillment, shipping updates, and returns |
| DTA-UC-05 | AI Marketing & Social Integration | Artisan, System | Auto-generate marketing posts and track campaign performance |
| DTA-UC-06 | Virtual Workshop / Training | Artisan, Trainer | Support artisans with business training events |
| DTA-UC-07 | Sales Analytics & Recommendations | Artisan, System | Provide analytics and AI suggestions for improvements |

| **Use Case ID** | **DTA-UC-01** |
|---|---|
| **Preconditions** | Artisan has internet access and required identity/payment docs. |
| **Trigger** | Artisan signs up to open a storefront. |
| **Main Flow** | 1. Artisan completes registration (personal and business details), uploads verification documents, selects storefront template, sets shop policies (return, shipping regions), and connects a bank or payment account.<br>2. The system validates documents (automated checks and manual review if needed), provisions the storefront, and provides a walkthrough for uploading products and marketing tools.<br>3. On successful activation, the storefront is indexed for search. |
| **Alternate Flows** | AF1: Verification fails — artisan guided to submit proof or request support. |
| **Postconditions** | Storefront created; artisan has access to dashboard. |
| **Special Requirements** | Secure payment onboarding; KYC processes as needed. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **DTA-UC-02** |
|---|---|
| **Preconditions** | Artisan logged in; storefront active. |
| **Trigger** | Artisan chooses to add a product. |
| **Main Flow** | 1. Artisan uploads photos, description, price, and inventory.<br>2. System validates image quality, auto-tags categories, suggests pricing and keywords for SEO, and creates alternate image sizes.<br>3. Artisan previews listing and publishes. System verifies compliance with prohibited items list. |
| **Alternate Flows** | AF1: Prohibited item detected — listing blocked with explanation. |
| **Postconditions** | Product published; search index updated. |

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| **Special Requirements** | Auto-tag accuracy and manual override. |
| **Frequency** | High |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **DTA-UC-03** |
| **Preconditions** | Product available in catalog; payment gateway operational. |
| **Trigger** | Buyer initiates purchase flow. |
| **Main Flow** | 1. Buyer searches or browses, views product details, adds to cart, and checks out.<br>2. The system calculates shipping and taxes, prompts for address, and processes payment via selected gateway.<br>3. On success, order confirmation is sent to buyer and artisan; inventory adjusted and shipping workflow initiated. |
| **Alternate Flows** | AF1: Payment declined — system retries or prompts alternate payment.<br>AF2: Item becomes unavailable — buyer notified and offered alternatives. |
| **Postconditions** | Order placed and recorded; confirmations sent. |
| **Special Requirements** | PCI compliance; multi-currency support. |
| **Frequency** | High |
| **Priority** | High |

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| **Use Case ID** | DTA-UC-04 |
| **Preconditions** | Order confirmed. |
| **Trigger** | Artisan marks order as ready/ships item. |
| **Main Flow** | 1. Artisan prints packing slip, marks item as shipped in dashboard, and provides tracking info that is propagated to buyer.<br>2. The system updates order status, triggers shipping label generation (if integrated), and tracks delivery.<br>3. Returns follow defined policy: buyer requests return, artisan approves, system issues RMA and coordinate logistics. |
| **Alternate Flows** | AF1: Lost shipment — initiate investigation and refund/replacement workflow. |
| **Postconditions** | Delivery status updated; records retained. |
| **Special Requirements** | Integration with logistics APIs; shipping cost calculations. |
| **Frequency** | High |
| **Priority** | High |

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| **Use Case ID** | **DTA-UC-05** |
| **Preconditions** | Artisan has social media accounts connected (optional). |
| **Trigger** | Artisan opts into marketing suggestions or scheduling. |
| **Main Flow** | 1. System analyzes product catalog and sales trends, suggests promotional copy, creates social media post drafts, and schedules posts to connected platforms.<br>2. It measures engagement and suggests optimization (best posting times, hashtags).<br>3. Artisan reviews and approves posts before publishing. |
| **Alternate Flows** | AF1: API integration fails — system provides manual copy for artisan to post. |

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| **Postconditions** | Campaign scheduled/published; analytics tracked. |
| **Special Requirements** | Rate limits for social APIs; privacy-consent for images/stories. |
| **Frequency** | Medium |
| **Priority** | Medium |

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|---|---|
| **Use Case ID** | **DTA-UC-06** |
| **Preconditions** | Workshop scheduled; trainer available; participants registered. |
| **Trigger** | Workshop session begins. |
| **Main Flow** | 1. Trainer opens live session; artisans join via web or mobile.<br>2. The system supports screen sharing, live Q&A, and resource distribution.<br>3. Attendance is recorded and certificates issued upon completion.<br>4. Recording is archived for later viewing. |
| **Alternate Flows** | AF1: Trainer cancels — participants notified and rescheduled. |
| **Postconditions** | Attendance recorded; resources archived. |
| **Special Requirements** | Low-latency streaming; multi-language captions preferable. |
| **Frequency** | Low |
| **Priority** | Medium |

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| **Use Case ID** | **DTA-UC-07** |
| **Preconditions** | Sales data exists. |
| **Trigger** | Artisan opens analytics dashboard or periodic email sent. |
| **Main Flow** | 1. System aggregates sales, traffic, conversion metrics and surfaces anomalies (e.g., spike in views).<br>2. AI suggests actions (price change, bundle offer, promoted listing).<br>3. The artisan can accept suggestions, schedule actions, or export reports. |
| **Alternate Flows** | AF1: Insufficient data — system suggests wait period to gather signals. |
| **Postconditions** | Recommendations executed or stored; reports exported if requested. |
| **Special Requirements** | Data privacy and anonymisation for marketplace-wide trends. |
| **Frequency** | Medium |
| **Priority** | Medium |

## C6: USE CASE MODEL FOR AI-POWERED PERSONALIZED CAREER GUIDANCE

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| PCG-UC-01 | Create Profile & Upload Portfolio | User (Student), System | Capture education, experience, skills and artifacts |
| PCG-UC-02 | Complete Skills & Interest Assessment | User, Assessment Engine | Generate inputs for career matching |
| PCG-UC-03 | Receive Career Recommendations | User, AI Career Module | Match user to career paths and courses |
| PCG-UC-04 | Enroll in Recommended Learning | User, Learning Platform | Register for courses to close skill gaps |
| PCG-UC-05 | Mentor Matching & Scheduling | User, Mentor, Counsellor | Pair user with mentor and schedule sessions |
| PCG-UC-06 | Job / Internship Matching & Application | User, Recruiter | Suggest job opportunities and assist with applications |
| PCG-UC-07 | Counsellor Review & Manual Adjustment | Counsellor | Human-in-the-loop validation and guidance |

| **Use Case ID** | **PCG-UC-01** |
|---|---|
| **Preconditions** | User has account or signs up. |
| **Trigger** | User selects "Create Profile" or "Upload Portfolio." |
| **Main Flow** | 1. User fills education, experience and skill fields, uploads CV/portfolio items (projects, code, demo reels).<br>2. System parses documents to extract structured skills (with NLP) and suggests additional tags.<br>3. The user verifies parsed skills and privacy settings, then saves the profile. |
| **Alternate Flows** | AF1: Parsing errors — user corrects fields manually. |
| **Postconditions** | Profile created and ready for assessment. |
| **Special Requirements** | Secure file storage; parsing accuracy and manual override. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **PCG-UC-02** |
|---|---|
| **Preconditions** | Profile exists. |
| **Trigger** | User starts assessment or scheduled periodic re-assessment. |
| **Main Flow** | 1. The assessment engine presents scenario-based tasks, situational judgement tests, or technical tasks.<br>2. The system evaluates responses and scores competencies, combining self-reported items with observed performance from portfolio and external credentials.<br>3. Results are stored and used as input to the matching engine. |
| **Alternate Flows** | AF1: User opts out of a section — system notes missing data and proceeds with partial scoring. |
| **Postconditions** | Skill profile updated; scores stored. |

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| **Special Requirements** | Assessment fairness; accommodations for disabilities. |
| **Frequency** | Medium |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **PCG-UC-03** |
| **Preconditions** | Assessment results and profile data available; job market data integrated. |
| **Trigger** | System runs matching algorithm or user requests suggestions. |
| **Main Flow** | 1. AI analyzes user data and labor market signals, producing a ranked list of career matches with required skills, expected salary, and recommended learning path.<br>2. Each recommendation includes rationale and links to relevant courses and job postings.<br>3. User can explore details, bookmark, or request counsellor review. |
| **Alternate Flows** | AF1: Low confidence in recommendation — system includes caveats and suggests exploratory options. |
| **Postconditions** | Recommendations saved; follow-up actions can be scheduled. |
| **Special Requirements** | Explainability of AI rationale; live labor market integration. |
| **Frequency** | Medium |
| **Priority** | High |

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| **Use Case ID** | **PCG-UC-04** |
| **Preconditions** | Course providers integrated; user account linked. |
| **Trigger** | User selects a recommended course. |
| **Main Flow** | 1. System shows course details, cost, duration and provider.<br>2. User enrolls (free or paid), and the course is added to their roadmap.<br>3. Completion updates skill profile, and the AI recalculates career fit. |
| **Alternate Flows** | AF1: Course full or unavailable — system suggests alternatives. |
| **Postconditions** | Enrollment recorded; roadmap updated. |
| **Special Requirements** | Single-sign-on with providers and certificate ingestion. |
| **Frequency** | Medium |
| **Priority** | Medium |

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|---|---|
| **Use Case ID** | **PCG-UC-05** |
| **Preconditions** | Mentor pool available; mentee preferences provided. |
| **Trigger** | User requests mentorship or system suggests a match. |
| **Main Flow** | 1. System runs compatibility scoring (skills, interest, availability) and suggests mentors.<br>2. User can request mentorship; mentor reviews candidate and accepts or declines.<br>3. Upon acceptance, the system facilitates scheduling, recurring sessions, and shared goals.<br>4. Mentors submit feedback and endorsements that update user profile. |
| **Alternate Flows** | AF1: Mentor declines — system suggests alternates. |
| **Postconditions** | Mentor-mentee relationship established; sessions scheduled. |

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| **Special Requirements** | Privacy of mentor/mentee contact details; scheduling integration. |
| **Frequency** | Low |
| **Priority** | Medium |

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|---|---|
| **Use Case ID** | **PCG-UC-06** |
| **Preconditions** | User profile and resume up-to-date; job feed active. |
| **Trigger** | System finds matching openings or user searches. |
| **Main Flow** | 1. System filters job postings by fit score and recommends roles.<br>2. User views job description and can apply through in-platform application (prefilled with profile and portfolio).<br>3. Recruiter receives candidate package and can rate or interview.<br>4. Interview scheduling and follow-up messages are handled within the platform. |
| **Alternate Flows** | AF1: Application requires external portal — system redirects and records the action. |
| **Postconditions** | Application recorded; recruiter notified. |
| **Special Requirements** | Data sharing consent with recruiters; anonymised pre-screening options. |
| **Frequency** | Medium |
| **Priority** | High |

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|---|---|
| **Use Case ID** | **PCG-UC-07** |
| **Preconditions** | Counsellor assigned or requested by user. |
| **Trigger** | Counsellor opens user case or scheduled review. |
| **Main Flow** | 1. Counsellor reviews AI recommendations and assessments, adjusts career plan, adds notes, and may schedule sessions.<br>2. Counsellor can override AI matches and add bespoke guidance; changes are logged and communicated to the user. |
| **Alternate Flows** | AF1: Counsellor requests additional evidence — system prompts user for verification. |
| **Postconditions** | Career plan updated; audit logs saved. |
| **Special Requirements** | Role-based access controls; auditability. |
| **Frequency** | Low |
| **Priority** | Medium |

## C7: USE CASE MODEL FOR GAMIFIED REHABILITATION FOR STROKE PATIENTS

| ID | USE CASE NAME | PRIMARY ACTORS | PURPOSE |
|---|---|---|---|
| GR-UC-01 | Patient Onboarding & Baseline Assessment | Physiotherapist, Patient, System | Capture baseline mobility and medical constraints |
| GR-UC-02 | Perform Guided Exercise Session (Motion-Tracked) | Patient, System, Wearable/Camera | Run a gamified exercise and capture motion data |
| GR-UC-03 | Therapist Assign / Adjust Exercise Plan | Physiotherapist, Patient | Create and tune therapy routines |
| GR-UC-04 | Real-Time Feedback & Safety Pause | System, Patient | Detect incorrect movement or fatigue and act |
| GR-UC-05 | Progress Monitoring & Reports | Physiotherapist, Caregiver, Patient | Track metrics, trends and adherence |
| GR-UC-06 | Remote Therapy / Tele-Session | Physiotherapist, Patient | Conduct live therapy with session recording |
| GR-UC-07 | Rewarding & Social Motivation | Patient, System, Peers | Gamify progress and enable safe social features |

| **Use Case ID** | **GR-UC-01** |
|---|---|
| **Preconditions** | Referral to program; patient consent and device availability. |
| **Trigger** | Therapist registers new patient on the platform. |
| **Main Flow** | 1. Therapist creates a patient profile, enters medical constraints and baseline assessments (range of motion, strength, cognitive notes).<br>2. The system suggests an initial exercise plan tailored by limb, side affected, and severity.<br>3. The patient receives an invitation to install the app, completes device calibration (camera/wearable), performs guided baseline tasks while system records metrics.<br>4. Therapist reviews baseline reports and finalizes the plan. |
| **Alternate Flows** | AF1: Device calibration fails — manual guidance provided or wearable recommended. |
| **Postconditions** | Baseline metrics stored; initial therapy plan activated. |
| **Special Requirements** | Medical data encryption; clinician sign-off on plan. |
| **Frequency** | Low |
| **Priority** | High |

| **Use Case ID** | **GR-UC-02** |
|---|---|
| **Preconditions** | Patient authenticated, device calibrated, exercise assigned. |
| **Trigger** | Patient begins scheduled session or starts ad-hoc exercise. |
| **Main Flow** | 1. The app starts the guided exercise, demonstrating movements with visuals and audio.<br>2. Camera/wearable tracks motion; the AI evaluates range, speed, and smoothness.<br>3. The patient receives real-time corrective cues (e.g., "lift your arm higher") and immediate gamified rewards for correct patterns.<br>4. Data are stored as session logs and aggregated for trend analysis.<br>5. If sensors detect potential safety concerns (e.g., sudden dizziness signals), the system pauses and prompts safety checks. |

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| **Alternate Flows** | AF1: Sensor dropout — system prompts reconnection and caches partial data.<br>AF2: Patient requests help — virtual assist or call-to-therapist initiated. |
| **Postconditions** | Session metrics stored; achievements updated; therapist notified for significant anomalies. |
| **Special Requirements** | Low-latency feedback; sensor fusion algorithms and safety thresholds. |
| **Frequency** | High |
| **Priority** | High |

| **Use Case ID** | **GR-UC-03** |
|---|---|
| **Preconditions** | Existing patient profile and session history. |
| **Trigger** | Periodic review or triggered by performance anomalies. |
| **Main Flow** | 1. Therapist reviews session history and analytics, adjusts exercise type / intensity / duration, and schedules new sessions.<br>2. Changes are pushed to the patient's app with explanations and demo videos.<br>3. The system validates changes against medical safety templates. |
| **Alternate Flows** | AF1: Therapist prescribes advanced wearable — system prompts procurement process. |
| **Postconditions** | Plan updated and synchronized. |
| **Special Requirements** | Clinical safety checks; role-based approvals for high-risk changes. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **GR-UC-04** |
|---|---|
| **Preconditions** | Session active and sensors functioning. |
| **Trigger** | Detection of incorrect movement pattern, excessive heart rate, or sudden instability. |
| **Main Flow** | 1. System flags anomaly, issues an audible/visual warning, pauses exercise, and provides recovery guidance (breathing, rest).<br>2. If critical safety thresholds are exceeded, it prompts the patient to call for help or offers to connect caregiver/therapist.<br>3. Events are logged and highlighted in the therapist dashboard. |
| **Alternate Flows** | AF1: False positive detection — patient can mark as false alarm; system uses feedback to refine thresholds. |
| **Postconditions** | Session paused and incident logged; therapist alerted if required. |
| **Special Requirements** | Fail-safe behavior; rapid notification channels. |
| **Frequency** | Low |
| **Priority** | Critical |

| **Use Case ID** | **GR-UC-05** |
|---|---|
| **Preconditions** | Session history exists. |
| **Trigger** | Therapist or caregiver requests a report; scheduled reporting period. |
| **Main Flow** | 1. The system aggregates session metrics, computes trend lines (accuracy, range, adherence), and generates reports with visualizations.<br>2. Reports highlight improvements, plateaus, and anomalies and suggest next steps (increase repetitions, add new exercises).<br>3. Reports can be exported for clinical records. |

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| **Alternate Flows** | AF1: Data gaps — system annotates missing days and recommends adjustments. |
| **Postconditions** | Report generated and shared with stakeholders per consent. |
| **Special Requirements** | Data both clinically meaningful and patient-friendly; export in standard formats (PDF/CSV). |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **GR-UC-06** |
|---|---|
| **Preconditions** | Teleconferencing module enabled; appointment booked. |
| **Trigger** | Scheduled remote session or urgent consult. |
| **Main Flow** | 1. Therapist initiates a live session with the patient, both join via secure audio/video channel with synchronized playback of exercise demos.<br>2. The therapist can observe motion-tracked metrics live, annotate, and guide the patient; session may be recorded with consent.<br>3. After the session, the therapist updates the plan and adds notes. |
| **Alternate Flows** | AF1: Connection degrades — system lowers video quality and prioritizes audio + sensor data. |
| **Postconditions** | Session recorded (if consented) and notes saved. |
| **Special Requirements** | HIPAA/PDPA-level security for streaming; recording consent management. |
| **Frequency** | Medium |
| **Priority** | High |

| **Use Case ID** | **GR-UC-07** |
|---|---|
| **Preconditions** | Gamification enabled; user consents to social features. |
| **Trigger** | Completion of sessions or milestones. |
| **Main Flow** | 1. Upon reaching milestones, the patient is awarded points/badges.<br>2. The system optionally shares progress to a closed group of peers or caregiver (with consent) to encourage social motivation.<br>3. Patients can join team challenges (step-count, adherence streaks).<br>4. Privacy controls allow limiting visibility. |
| **Alternate Flows** | AF1: Patient opts out of social sharing — all sharing disabled. |
| **Postconditions** | Rewards recorded; social feed updated if permitted. |
| **Special Requirements** | Prevent competitive pressure; focus on rehabilitation goals. |
| **Frequency** | Medium |
| **Priority** | Medium |